Machine Learning for Traffic Flow Prediction in Urban Areas

UE24CS352A — Machine Learning Mini-Project

1. Problem Statement

Urban congestion causes delays, wasted fuel, and uncertainty for commuters. This project formulates traffic prediction as a five-class classification task: estimate the congestion level for a road using its identifier, day of week, and hour of day. The label scale follows the supplied reference paper: 0 represents very low congestion and 4 represents very high congestion.

2. Dataset Details

The supplied paper describes data obtained through OutScraper and discusses 35 streets around Central Park, New York, with road, day, and hour features. The raw source dataset was not included in the provided files. For reproducible software testing, this project therefore includes a synthetic demonstration CSV with the same four-column schema: road_id, day_of_week, hour, congestion_level. It is generated to resemble weekday commute peaks and is not a substitute for measured traffic data. Any performance computed on it must be labelled synthetic-demo performance.

3. Approach

The pipeline validates the required columns, removes invalid or duplicate records, and splits the data into training and test sets using stratification. It trains (a) a Multi-Layer Perceptron (MLP) neural network with one hidden layer and five output classes and (b) a Random Forest baseline. Models are compared using accuracy and macro-F1. Confusion matrices and per-class precision, recall, and F1 reports are saved. Macro-F1 is included because it gives each congestion class equal weight.

4. Implementation Overview

The implementation is organized into data validation (src/data_utils.py), model training and evaluation (src/train.py), command-line prediction (src/predict.py), and an optional Streamlit interface (src/app.py). The selected model is saved as outputs/model.joblib. The README contains macOS setup, training, prediction, and real-data replacement instructions.

5. Evaluation and Results

Run `python src/train.py` to calculate the actual held-out metrics for the selected CSV. The report deliberately does not claim a real-world accuracy value because the original measured dataset was not supplied. Results produced from traffic_demo.csv are software demonstration results only; a valid real-world claim requires a documented real dataset and an appropriate test split.

6. Expected Demonstration

During the review, demonstrate data validation, model training, comparison of the MLP and Random Forest, the saved confusion matrices, and a prediction for a selected road/day/hour. The optional dashboard accepts the same three inputs and displays the predicted class and class probabilities when supported by the model.

7. Limitations

The feature set is intentionally small and does not include real-time vehicle counts, speed sensors, weather, incidents, holidays, road capacity, or explicit spatial relationships between roads. The synthetic dataset encodes simplified patterns and can make the task easier or different from reality. Random train/test splitting may also overestimate performance for future-time prediction; a real deployment should use chronological validation and compare predictions against measured outcomes.

8. Conclusion and Future Work

This project provides an end-to-end baseline for congestion-level classification, including data checks, two models, evaluation artifacts, a prediction command, and a small interactive dashboard. The next step is to obtain an authorized measured traffic dataset, document its provenance and collection period, use time-aware evaluation, and extend the feature set with traffic speed/volume, weather, incidents, and neighboring-road information. Sequence models such as LSTM can be investigated if sufficient timestamped observations are available.

Reference

Huang, Y. and Zhou, G. (2021). “Machine Learning for Traffic Flow Prediction in Urban Areas.” Supplied reference paper. Dataset description and baseline design are summarized from the provided PDF; the original raw dataset was not provided with the assignment.